

SentinelEdgeAI 3D Enclosure — STL-Ready Dimension Specification

Overall Shell

Parameter	Value	Notes
Length (X)	220.0 mm	Front-to-back
Width (Y)	140.0 mm	Left-to-right
Height (Z)	60.0 mm	Bottom-to-top
Wall thickness	2.5 mm	Aluminum sheet / ABS shell
Corner radius	8.0 mm	All outer vertical edges
Top corner radius	4.0 mm	Top face inner fillet

Top Panel

Feature	X (mm)	Y (mm)	W (mm)	H (mm)	Notes
LED screen cutout	30	22	160	95	Centered; recessed 2 mm
Screen mount holes	32/186	24/115	—	—	M3 × 4 corners, Ø 3.2 mm
Rear vent strip × 3	170	5	40	4	Spaced 6 mm apart

Front Panel

Feature	X (mm)	Y (mm)	W (mm)	H (mm)	Notes
Intake fan mount	10	10	40	40	40×40 mm fan, M3 × 4 holes
Fan hole (Ø 37 mm)	11.5	11.5	37 dia	—	Circular cutout
Vent grille strips	58	8	50	4	× 8 strips, 6 mm spacing

Feature	X (mm)	Y (mm)	W (mm)	H (mm)	Notes
Status LED row	115	48	5 dia	—	× 4 LEDs, 6 mm apart, Ø 5

Rear Panel

Feature	X (mm)	Y (mm)	W (mm)	H (mm)	Notes
Exhaust fan mount	90	10	40	40	40×40 mm fan, M3 × 4 holes
Fan hole (Ø 37 mm)	91.5	11.5	37 dia	—	Circular cutout
Vent slots × 4	140	10	30	6	8 mm spacing

Right-Side Port Panel

All ports flush-mounted, single face. Left edge = front of enclosure.

Port	X offset (mm)	Z offset from bottom (mm)	Cutout W × H (mm)	Notes
WAN (RJ45)	12	12	16 × 14	Ethernet female
LAN (RJ45)	34	12	16 × 14	Ethernet female
USB-A × 2	58	14	14 × 6.5 each	4 mm gap between
HDMI	88	13	15 × 8	HDMI Type A
DC Power	112	16	Ø 8.0 mm	5.5/2.1 mm barrel

Internal Component Mounts (standoffs)

All standoffs M3, height 5 mm above floor.

Jetson Orin Nano

- Module size: 69.6×45 mm
- Mount position (X,Y from front-left inside corner): X=15, Y=20
- Mounting holes: $4 \times$ M3, per NVIDIA Jetson Orin Nano carrier spec
- Clearance above: 25 mm minimum

Raspberry Pi 5

- Module size: 85×56 mm
- Mount position: X=100, Y=20
- Mounting holes: $4 \times$ M3, standard Pi HAT pattern (58×49 mm)
- Clearance above: 20 mm minimum

M.2 SSD Bracket

- SSD size: M.2 2280 (80×22 mm)
 - Mount: side-wall bracket at Y=118, X=20-100
 - Bracket height from floor: 8 mm
 - Single M2 screw retention
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Airflow Path

[FRONT] --> Intake fan (40mm) --> Jetson --> Pi 5 --> Exhaust fan (40mm) --> [REAR]

- Airflow direction: +X (front to back)
 - Minimum channel clearance between boards and shell: 8 mm sides, 10 mm top
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Cable Routing

- Internal zip-tie anchor points: $6 \times$ (2 on floor, 2 on sidewalls, 2 on top underside)
 - Grommet slot for internal ribbon/power: X=10, Y=70, Z=floor, W=20, H=6 mm
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Bill of Materials (hardware)

Item	Qty	Spec
M3 × 6 mm screws	20	Pan head
M3 × 10 mm screws	8	For fan mounts
M3 brass standoffs	12	5 mm height
40mm × 10mm fan	2	5V DC, 2-pin
M2 screw (SSD)	1	M2 × 3 mm
Rubber feet	4	Ø 10 mm × 3 mm

CAD Prompt (paste into Fusion 360 / FreeCAD AI assistant)

Create a rectangular enclosure: 220mm L × 140mm W × 60mm H

- 2.5mm wall thickness, 8mm outer corner radius
- Top face: recessed 160×95mm cutout for 7-inch display, centered at (30,22) from top-left
- Front face: 40mm fan mount at (10,10) with M3 holes, Ø37mm fan hole
- Rear face: 40mm fan mount at (90,10) with M3 holes, Ø37mm fan hole
- Right face: port cutouts at X=12,34,58,88,112 for RJ45×2, USB-A×2, HDMI, DC barrel
- Floor: M3 standoffs at Jetson Orin Nano positions (X15,Y20) and Pi5 positions (X100,Y20)
- Rear vent strip: 3× 40×4mm slots at top-rear
- Material: aluminum 6061 or ABS+
- Include snap-fit or 4× M3 screw lid retention

Recommended Slicer Settings (FDM / ABS+)

Parameter	Value
Layer height	0.2 mm
Infill	30% gyroid

Parameter	Value
Perimeters	4
Support	Everywhere
Bed temp	100°C
Nozzle temp	240°C
Material	ABS+ or ASA

Generated for SentinelEdgeAI v1.0 — ready for Fusion 360, FreeCAD, OpenSCAD, or Bambu/Prusa slicer import.